Figure 4: HR

a new Job opening in the portal, we can also share the information about the job.

The ERPNext Web portal gives one's customers quick access to their Orders, Invoices and Shipments. Customers can check the status of their orders, invoices and shipping status by logging on to the Web. To log into their account and check the order status, customers have to use their email ID and the password sent by ERPNext, generated through the sign-up process. The best part is that ERPNext provides Web support, i.e., static content like Home page, About Us and Contacts can be created using the Web page by going to *Website-> Web Page->New*.

ERPNext also offers customisation. One can simplify the forms by hiding features not needed, by using *Disable Features, Module Setup, Add Custom Fields*. One can also change form properties, like adding more options to drop-down boxes or hiding fields using *Customize Form View*, and make one's own Print Formats by using HTML Templates.

ERPNext is a very well integrated tool, which caters to almost all the requirements of a company and especially a start-up. I would recommend it for all those young entrepreneurs who have set up a company and will be working hard to fulfill their dreams. END

By: Ashish Kumar Sinha

The author is a software engineer based in Bengaluru. A software enthusiast by heart, he is passionate about using open source technology and sharing it with the world. He can be reached at *ashi.sinha.87@gmail.com*

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```
$etmp4 -f -0 -c Send_time_file Recv_time_file Source_Video_
Trace bus_cif2.mp4 bus_cif_recv.mp4
```

This generates a (possibly corrupted) video file, in which all frames that got lost or were corrupted are deleted from the original video track. The reconstructed video will also demonstrate the effect of jitter introduced during packet transmission.

etmp4 also creates some other files, which are listed below:

loss_bus_cif_recv.txt contains I, P, B and overall frame loss in percentage.

delay_bus_cif_recv.txt contains frame-nr., lost-flag, end-to-end delay, inter-frame gap sender, inter-frame gap receiver, and cumulative jitter in seconds.

rate_s_bus_cif_recv.txt contains time, bytes per second (current time interval), and bytes per second (cumulative) measured at the sender and receiver.

The PSNR by itself does not mean much; other quality metrics that calculate the difference between the quality of the encoded video and the received video (which could be corrupted) can be used. Using a specialised tool like MSU, video quality in terms of VQM, SSIM can be measured, as transmitted and received videos are available.

7. Now, decode the received video to *yuv* format:

```
$ffmpeg -i bus_cif_recv.mp4 bus_cif_recv.yuv
```

8. Compute the PSNR:

```
psnr x y <YUV format> <src.yuv> <dst.yuv> [multiplex] [ssim]
```

```
$psnr 352 288 420 bus_cif2.yuv bus_cif_recv.yuv > ref_psnr.txt
```

Simulating an experimental set-up

All relevant files for simulation can be downloaded from *http://opensourceforu.efytimes.com/article_source_code/july15/vt_ns2.zip*.

Performance evaluation

We can plot the graph PSNR as mentioned in Step 8. We can also plot other quality metrics like VQM and SSIM. END

References

[1] Klaue, Jirka, Berthold Rathke, and Adam Wolisz;'Evalvid–A framework for video transmission and quality evaluation. Computer Performance Evaluation. Modelling Techniques and Tools". Springer Berlin Heidelberg, 2003. 255-272.

[2] *http://www2.tkn.tu-berlin.de/research/evalvid/EvalVid/docevalvid.html* source file.

By: Vijay Ukani and Jitendra Bhatia

Vijay Ukani is associate professor at Institute of Technology, Nirma University, Ahmedabad. He enjoys teaching and exploring network protocols and multimedia transmission. Connect with him at *vijayukani@yahoo.com*.

Bhatia is senior assistant professor at the Department of Computer Science and Engineering, Institute of Technology, Nirma University, Ahmedabad. His research interests are in the area of computer networks and software defined networking. Connect with him at *jitendrabbhatia@gmail.com*.